

THIMIRA BANDARA

Data Engineer

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SUMMARY

- Data Engineer specializing in scalable ETL pipelines, RESTful Python workflows, and big data systems like Spark and Databricks, with a strong focus on collaborative infrastructure development to support data-driven decision-making.

EXPERIENCE

Data Engineer

City of Aurora - IT

📅 Feb 2024 – Present 📍 Aurora, IL

- Developed scalable, end-to-end ETL pipelines using REST APIs and Python, orchestrated via Azure Data Factory and deployed through CI/CD to automate municipal data ingestion.
- Led a legacy-to-cloud migration using distributed Python workflows to extract, map, and validate data—centralizing departmental systems and improving access for better cross-team insights.
- Implemented a secure RSA-encrypted linked service with subnet-restricted network configurations to ingest sensitive financial and PII data from a Snowflake server into a data lake.
- Translated complex data workflows and schemas for city stakeholders and community leaders, ensuring clear communication across technical and non-technical teams to support informed decision-making across departments.

Data Engineer

Nirvana Labs

📅 May 2022 – Dec 2023 📍 Indianapolis, IN

- Built and deployed a high-speed News API using Node.js and PostgreSQL, scraping and aggregating 10K+ financial articles to generate domain-specific training data for FinBERT-based sentiment modeling.
- Developed a custom KNN-based classification algorithm to identify behavioral patterns in blockchain wallets, supporting fraud detection and investor segmentation efforts.
- Automated model-ready sentiment processing with Python scripts and Cron scheduling, reducing latency and ensuring fresh data delivery to live dashboards and ML endpoints.
- Partnered with data scientists and product teams to define schema requirements and optimize NLP pipeline outputs for downstream analytics and internal reporting tools.

EDUCATION

University of Illinois Urbana-Champaign

M.S. in Computer Science GPA: 3.9/4.0 (2022–2024)

B.S. in Earth, Society, and Environmental Sustainability GPA: 3.5/4.0 (2018–2022)

SKILLS

Programming & Tools

Python SQL Node.js Git Linux .NET Framework REST APIs FastAPI

Data Engineering

ETL Pipelines Spark Databricks Airflow PostgreSQL NoSQL Snowflake Data Lakes

Cloud & DevOps

Azure AWS Docker Kubernetes CI/CD

PROJECTS

Energy + Renewables Data Pipeline

Personal Project

📅 2024

- Built a data pipeline to explore patterns between energy demand and renewable output, using REST APIs for extraction, Data Factory for ETL, Databricks for processing, and Power BI for visualization.

Facebook Marketplace Car Finder Bot

Personal Project

📅 2024

- Created a Discord-integrated vehicle search assistant using Selenium to scrape Facebook Marketplace listings based on city, mileage, make, year, and price filters.
- Developed Python-based logic to rank and return the top 10 most relevant listings, delivering them directly to users via a custom Discord bot interface.
- Optimized scraper behavior to mimic human interaction, bypass rate limits, and maintain uptime across multiple search requests in real-time.

Graduate Research Match Platform

Academic Project

📅 2023

- Designed a web-based app to connect students with graduate research opportunities by matching academic interests with relevant programs, institutions, and faculty across the U.S.
- Aggregated and normalized data from multiple sources (university directories, publication databases, and ranking systems) to build a searchable network of schools and advisors.
- Implemented advanced filtering and ranking logic to surface personalized recommendations based on interests, research areas, and future career goals.
- Enabled real-time search and save functionality using FastAPI and PostgreSQL, improving accessibility to research pathways for undergraduates exploring graduate options.